Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression

IDE: Google Colab

Theory:

Linear regression is a method for determining the best linear relationship between two variables *X* and *Y*. If variables *X* and *Y* are uncorrelated, it is pointless embarking upon linear regression. However, if a reasonable degree of correlation exists between *X* and *Y* then linear regression may be a useful means to describe the relationship between the two variables. The usual approach is to use the *least-squares* method, which minimizes the squared difference between the actual data points and a straight line. Let $[x_i, y_i]$, $i = 1,2,3,....,N$ be the *N* pairs of data values of the variables *X* and *Y*. The straight-line relating *X* and *Y* is $y = mx + c$, where *m* and *c* are the gradient and constant values (to be determined) defining the straight line. Thus, $y(x_i) - y_i$ is the difference between the line and data point *i* (see Fig. 1). Taking all the data points, we seek values of *m* and *c* that minimize the squared difference *SD*.

$$\sum_{1}^{N}\left[y(x_i) - y_i\right]^2$$

This is achieved by calculating the partial derivatives of *SD* with respect to *m* and *c* and finding the pair [*m*,*c*] such that *SD* is at a minimum.

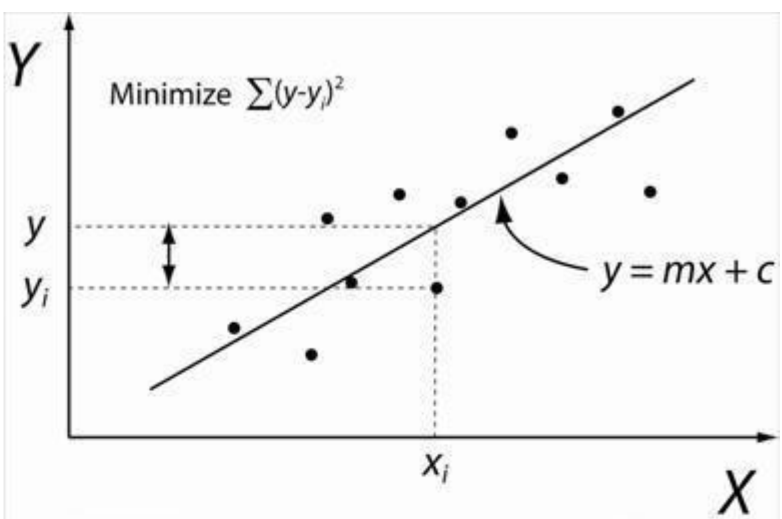


Figure 1: Illustration of Linear Regression. Linear least squares regression, the idea is to find the line y = mx + c that minimizes the mean squared difference between the line and the data points

Batch Gradient Descent:

Gradient Descent is an optimization algorithm used for minimizing the cost function in various machine learning algorithms. It is basically used for updating the parameters of the learning model. Batch gradient descent which processes all the training examples for each iteration of gradient descent. But if the number of training examples is large, then batch gradient descent is computationally very expensive.

Let m be the number of training examples. Let n be the number of features.

Algorithm for batch gradient descent :

Let $h_\theta(x)$ be the hypothesis for linear regression. Then, the cost function is given by:
Let Σ represents the sum of all training examples from i=1 to m.
$J_{train}(\theta) = (1/2m)\ \Sigma(\ h_\theta(x^{(i)})\ -\ y^{(i)})^2$

Repeat {
$\theta j = \theta j$ – (learning rate/m) * $\Sigma(\ h_\theta(x^{(i)})\ -\ y^{(i)})x_j^{(i)}$
For every j =0 …n
}
Where $x_j^{(i)}$ Represents the j^{th} feature of the i^{th} training example. So if *m* is very large(e.g. 5 million training samples), then it takes hours or even days to converge to the global minimum.That's why for large datasets, it is not recommended to use batch gradient descent as it slows down the learning.

Pre Lab Exercise:

a. Explain the meaning of linear regression
Linear regression is a method for determining the best linear relationship between two variables X and Y. If variables X and Y are uncorrelated, it is pointless embarking upon linear regression.

b. Write three applications of linear regression
- Stock Market Trend Prediction
- Weather Parameter Estimation
- Demand Forecasting in Supply Chain Management

c. Write three advantages of linear regression
   - Simple and Easy to understand
   - Fast to train and interpret
   - Works well when relationship between variables is approximately linear

d. Write three limitations of linear regression
   - Fails for non-linear data
   - Sensitive to Outliers
   - Performance drops with multicollinearity and irrelevant features

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Pre-process the data
5. Visualize the Data
6. Separate the feature and prediction value columns
7. Write the Hypothesis Function
8. Write the Cost Function
9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Find the best fit line to the given dataset
12. Observe the cost function vs iterations learning curve

Program (Code):

Simple Linear Regression:

Code:

```
import numpy as np
import matplotlib.pyplot as plt

def estimate_coef(x, y):
# number of observations/points
  n = np.size(x)

# mean of x and y vector
  m_x = np.mean(x)
  m_y = np.mean(y)
```

```
# calculating cross-deviation and deviation about x
  SS_xy = np.sum(y*x) - n*m_y*m_x
  SS_xx = np.sum(x*x) - n*m_x*m_x

# calculating regression coefficients
  b_1 = SS_xy / SS_xx
  b_0 = m_y - b_1*m_x

  return (b_0, b_1)

def plot_regression_line(x, y, b):
# plotting the actual points as scatter plot
  plt.scatter(x, y, color = "m", marker = "o", s = 30)

# predicted response vector
  y_pred = b[0] + b[1]*x

# plotting the regression line
  plt.plot(x, y_pred, color = "g")

# putting labels
  plt.xlabel('x')
  plt.ylabel('y')
  plt.show()

def main():
  # observations / data
  x = np.array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
  y = np.array([1, 3, 2, 5, 7, 8, 8, 9, 10, 12])

  # estimating coefficients
  b = estimate_coef(x, y)
  print("Estimated coefficients:\nb_0 = {}\nb_1 = {}".format(b[0], b[1]))

  # plotting regression line
  plot_regression_line(x, y, b)

main()
```

Multiple Linear Regression:

```
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import numpy as np
from sklearn import datasets, linear_model, metrics
import pandas as pd

data_url = "https://lib.stat.cmu.edu/datasets/boston"
raw_df = pd.read_csv(data_url, sep=r"\s+", skiprows=22, header=None)

X = np.hstack([raw_df.values[::2, :], raw_df.values[1::2, :2]])
y = raw_df.values[1::2, 2]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=1)

reg = linear_model.LinearRegression()
reg.fit(X_train, y_train)

# regression coefficients
print('Coefficients: ', reg.coef_)

# variance score: 1 means perfect prediction
print('Variance score: {}'.format(reg.score(X_test, y_test)))

# plot for residual error

# setting plot style
plt.style.use('fivethirtyeight')

# plotting residual errors in training data
plt.scatter(reg.predict(X_train),
        reg.predict(X_train) - y_train,
        color="green", s=10,
        label='Train data')

# plotting residual errors in test data
plt.scatter(reg.predict(X_test),
        reg.predict(X_test) - y_test,
        color="blue", s=10,
        label='Test data')
```

```
# plotting line for zero residual error
plt.hlines(y=0, xmin=0, xmax=50, linewidth=2)

# plotting legend
plt.legend(loc='upper right')

# plot title
plt.title("Residual errors")

# method call for showing the plot
plt.show()
```

Polynomial Linear Regression

```
#Polynomial Linear Regression

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn
import warnings

from sklearn.preprocessing import LabelEncoder
from sklearn.impute import KNNImputer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import f1_score
from sklearn.ensemble import RandomForestRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import cross_val_score

warnings.filterwarnings('ignore')

df = pd.read_csv('Position_Salaries.csv')

X = df.iloc[:,1:2].values # Corrected to select 'Level' column as a 2D array
y = df.iloc[:,2].values  # Corrected to select 'Salary' column

from sklearn.linear_model import LinearRegression
lin_reg=LinearRegression()
lin_reg.fit(X,y)
```

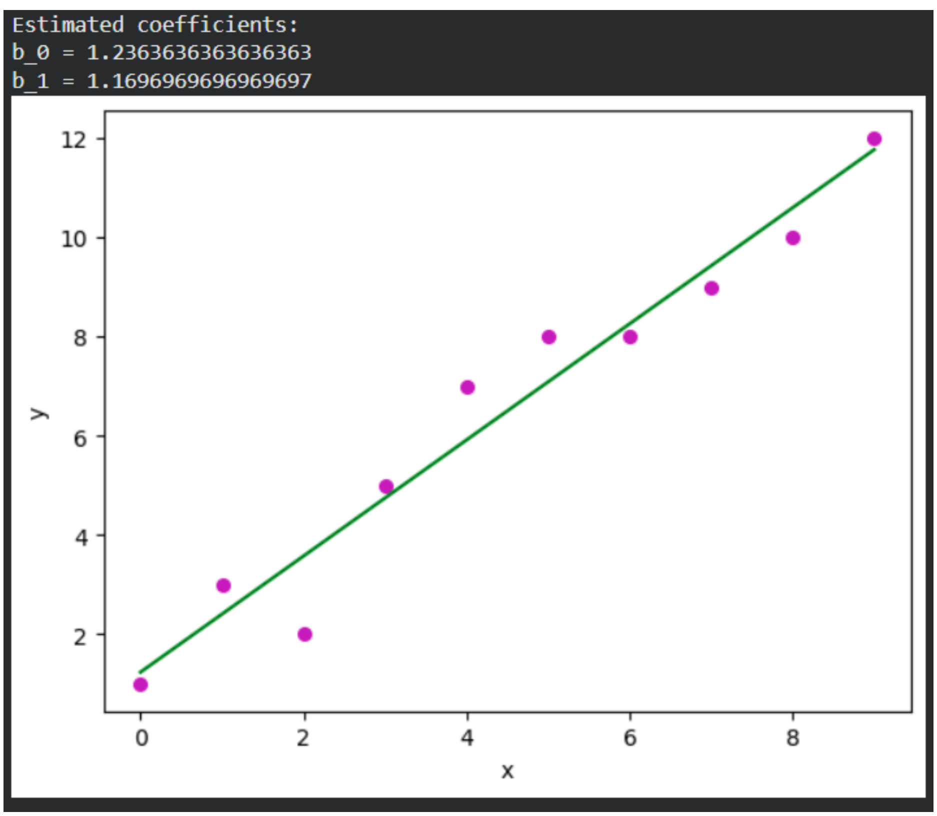
```
from sklearn.preprocessing import PolynomialFeatures
poly_reg2=PolynomialFeatures(degree=2)
X_poly=poly_reg2.fit_transform(X)
lin_reg_2=LinearRegression()
lin_reg_2.fit(X_poly,y)

poly_reg3=PolynomialFeatures(degree=3)
X_poly3=poly_reg3.fit_transform(X)
lin_reg_3=LinearRegression()
lin_reg_3.fit(X_poly3,y)

plt.scatter(X,y,color='red')
plt.plot(X,lin_reg.predict(X),color='green')
plt.title('Simple Linear Regression')
plt.xlabel('Position Level')
plt.ylabel('Salary')
plt.show()
```

Results:

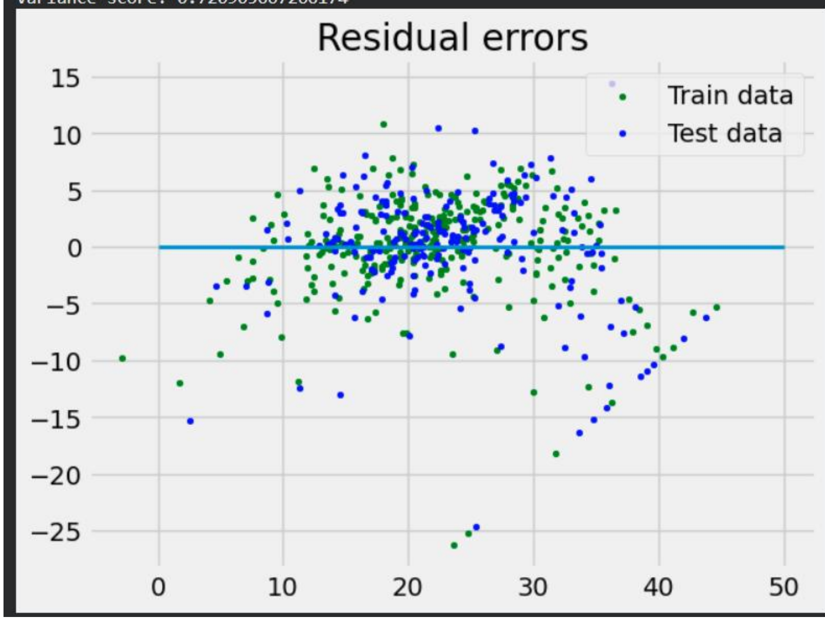
Simple Linear Regression:



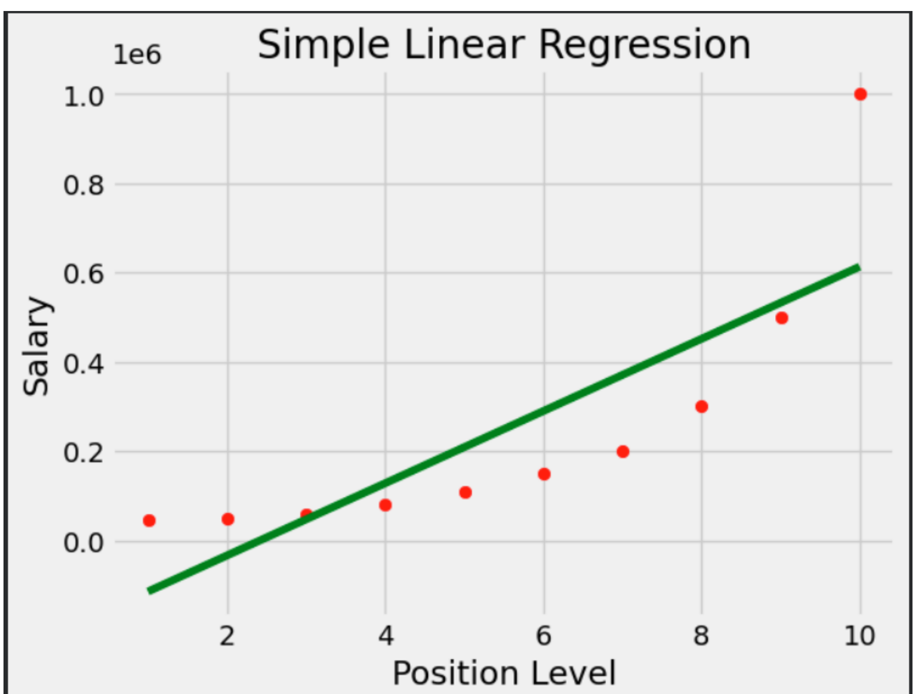
Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression	
Experiment No: 02	Date:	Enrollment No: 92301733043

Multiple Linear Regression:

```
Coefficients:  [-8.95714048e-02  6.73132853e-02  5.04649248e-02  2.18579583e+00
 -1.72053975e+01  3.63606995e+00  2.05579939e-03 -1.36602886e+00
  2.89576718e-01 -1.22700072e-02 -8.34881849e-01  9.40360790e-03
 -5.04008320e-01]
Variance score: 0.720905667266174
```



Polynomial Linear Regression:



Observation and Result Analysis:

a. Nature of the dataset

- Simple Linear Regression: small, univariate, continuous, labeled dataset.
- Multiple Linear Regression: multivariate, numerical, labeled, real-world dataset.
- Polynomial Linear Regession: small, univariate, numerical, labeled dataset

b. During Training Process

- Simple Linear Regression:
  - Supervised and labeled (each x has a y)
  - Numerical and continuous
  - Univariate (1 input, 1 output)
  - Structured/tabular
  - Used to learn a linear relationship

- Multiple Linear Regression:
  - Supervised and labeled
  - Numerical and continuous
  - Multivariate (many input features, one output)
  - Structured/tabular
  - Represents a linear relationship in higher dimensions

- Polynomial Linear Regression:
  - Supervised and labeled
  - Numerical and continuous
  - Univariate (in your case)
  - Structured/tabular
  - Non-linear in original form, but converted into polynomial features
  - Model still trained using linear regression on transformed data

c. After the training Process

- Simple Linear Regression:
  - The dataset is represented by a learned linear model:

$$y = b_0 + b_1 x$$

  - Data can now be:
    - Predicted dataset (ŷ values)
    - Residual dataset (errors = y − ŷ)

  - Nature: Numerical, continuous, model-fitted dataset used for prediction and evaluation.

- Multiple Linear Regression:

- The dataset is represented by a multivariate linear equation:

$$y = b_0 + b_1x_1 + b_2x_2 + \ldots + b_nx_n$$

- Dataset is split into:
  - Predicted values
  - Residuals/errors

- Nature: Multivariate numerical, continuous, trained regression output dataset.

- Polynomial Linear Regression:
  - Original data is transformed into polynomial features
  - The final dataset follows a non-linear fitted curve
  - Used to generate:
    - Predicted salaries
    - Error/residual values
  - Nature: Numerical, continuous, non-linearly fitted regression dataset.

Post Lab Exercise:

a. What are the major assumptions considered in linear regression
   - Linearity
   - Independence of Errors
   - Normality of error
   - No Multicollinearity
   - No or Little Outliers

b. Why MSE is used instead of MAE for calculating the loss function
   - Penalizes large errors more
     MSE squares the errors, so large mistakes get much higher penalty, which helps the model focus on reducing big errors.

   - Differentiable everywhere
     MSE is smooth and differentiable, making it easier to optimize using gradient descent.
     MAE is not differentiable at zero.

   - Gives a unique global minimum
     MSE produces a convex cost function, ensuring a single best solution for linear regression.

   - Mathematically convenient
     MSE leads to closed-form solutions (Normal Equation) and stable convergence.

c. How can the behavior of outliers be understood while dealing with the unseen dataset
   - Analyzing residuals (prediction errors)
     Large residuals on unseen data indicate possible outliers or poor generalization.

- Using visualization techniques
  Scatter plots, box plots, and residual plots help detect abnormal patterns or extreme values.

- Checking statistical measures
  Z-score, IQR, and leverage/Cook's distance can identify whether unseen points are influential outliers.

- Comparing train vs test performance
  If error suddenly increases on unseen data, it suggests outliers or distribution shift.

- Model robustness testing
  Observing how predictions change when extreme values appear helps understand outlier sensitivity.

d. Derive the Normal Equation for the Linear Regression.

31 DECEMBER
Sunday 2017

Linear Regression Model

- for multiple Linear Regression, the hypothesis is $h_\theta(x) = X\theta$

where,

X = design matrix
θ = parameter vector
y = output vector

- Cost function (MSE)

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}(X\theta - y)^2$$

In matrix form:

$$J(\theta) = \frac{1}{2m}(X\theta - y)^T(X\theta - y)$$

- Take partial derivative w.r.t. θ

$$\frac{\partial J(\theta)}{\partial \theta} = \frac{1}{m}X^T(X\theta - y)$$

- Set gradient to 0

$$X^T(X\theta - y) = 0$$
$$X^T X\theta = X^T y$$

- Solve for θ

$$\theta = (X^T X)^{-1} X^T y$$ ← Final eqn

NOV 2017
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